

Honors Analysis I MATH 20700



17 Oct 2025

→ Series $\sum_{n=1}^{\infty} a_n$ converges if the sequence
of partial sums $\sum_{n=1}^N a_n$ converges.

→ Geometric series $\sum_{n=1}^{\infty} \lambda^n$

if $|\lambda| \geq 1$
then diverges

if $|\lambda| < 1$ then
converges to
 $\frac{1 - \lambda^{N+1}}{1 - \lambda}$

Silly criterion (necessary but not
sufficient condition)

if $\sum_{n=1}^{\infty} a_n$ converges then $\lim_{n \rightarrow \infty} a_n = 0$.

If $b_n \geq 0$ and $\sum b_n$ converges
and $|a_n| < b_n$ then $\sum a_n$ converges

Def $\sum a_n$ converges absolutely if
 $\sum |a_n|$ converges.

IMP defn

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\sup_{m \geq n} a_m \right).$$

Root Test

If $\limsup_{n \rightarrow \infty} |a_n|^{1/n} < 1$ then $\sum a_n$ converges
 > 1 then diverges
 $= 1$ then inconclusive

Proof of Root Test

$$\text{Let } \lambda = \limsup_{n \rightarrow \infty} |a_n|^{1/n} < 1$$

By defn of $\limsup$, $\forall \varepsilon > 0$, $\exists N$ s.t.

$$\forall n > N, |a_n|^{1/n} < \lambda + \varepsilon.$$



$$|a_n| < (\lambda + \varepsilon)^n$$



Pick $\varepsilon > 0$ s.t. $\lambda + \varepsilon < 1$
then $|a_n| < \text{Geometric Series}$

Ratio Test

$$\text{Let } \lambda = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}$$

If $\lambda < 1$ then $\sum a_n$ converges.

If $\lambda > 1$ then diverges

If $\lambda = 1$ then inconclusive.

$$\text{If } \limsup_{n \rightarrow \infty} |a_n|^{1/n} = 1?$$

$$\sum \frac{1}{n} = +\infty$$

$$\left(\frac{1}{n}\right)^{1/n} \rightarrow 1 \text{ as } n \rightarrow \infty$$

$$\text{Since } \left(\frac{1}{n}\right)^{1/n} = \exp\left(-\frac{1}{n} \log n\right)$$

$\downarrow$
0

$\rightarrow 1.$

See that

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots$$

Integral Test

$$1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \dots$$

$$\downarrow$$
$$+\infty.$$

To prove that $\sum a_n < +\infty$
($a_n > 0$), we can use

Integral Test.

① Find $f(x)$ s.t. $f(x) \geq a_n$ for all
 $x \in [n-1, n]$

② Then if $\int_0^{\infty} f(x) dx$ is finite, we know
 $\sum a_n \leq \int f(x) dx$ is finite.

E.g. to check $\sum \frac{1}{n^2}$ converges,

$$\text{set } f(x) = \begin{cases} 1/x^2 & \text{if } x > 1 \\ 1 & \text{if } x < 1 \end{cases}$$

$$\begin{aligned} \text{Then } \int_0^{\infty} f(x) dx &= 1 + \int_1^{\infty} f(x) dx \\ &= 1 + \int_1^{\infty} \frac{1}{x^2} dx \end{aligned}$$

which is finite.

The series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$ converges

but does not converge absolutely
(it is said to converge conditionally)

Thm If $\sum_{n=1}^{\infty} a_n$ converges conditionally but

not absolutely, then $\forall \lambda \in \mathbb{R}, \exists$ rearrangement
that converges to λ .

Let a_n be a seq. so that

a_n alternates signs
and $|a_n|$ is decreasing
and $|a_n| \rightarrow 0$

Then $\sum_{n=1}^{\infty} a_n$ converges.

Alternating Series Test

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→ If $\sum_{k=1}^{\infty} a_k$ converges absolutely, then

for any permutation $p: \mathbb{N} \rightarrow \mathbb{N}$ bijective,

$\sum_{k=1}^{\infty} a_{p(k)}$ converges to the same limit

→ Integral, ratio, root test.

→ Check that root test implies ratio test (but not other way round)

→ Use ratio test to see for what values of x we have $\sum_{n=1}^{\infty} x^n \cdot \frac{n^n}{n!}$ convergent

→ Power series $\sum_{n=0}^{\infty} a_n x^n$

→ Proof Let $R = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{1/n}}$

$$= \liminf_{n \rightarrow \infty} |a_n|^{-1/n}$$

Then series converges absolutely for $|x| < R$ and diverges for $|x| > R$.

→ Proof Apply root test

$$\lambda = \limsup_{n \rightarrow \infty} |a_n x^n|^{1/n}$$
$$= |x| \limsup_{n \rightarrow \infty} |a_n|^{1/n} = \frac{|x|}{R}$$

So clearly we need $|x| < R$ for convergence

R is called radius of convergence

→ E.g. Taylor expansion for $\log$ is

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$$

then it has radius of convergence $R=1$.

→ Convergence of sequences of functions

Consider $f_n: M \rightarrow N$ with M, N metric spaces

① $f_n \rightarrow f$ converges pointwise if

$\forall x \in M, f_n(x) \rightarrow f(x)$ as a sequence in N

② $f_n \rightarrow f$ converges uniformly if $\forall \varepsilon > 0,$

$\exists N \in \mathbb{N}$ s.t. $\forall n > N, \forall x \in M,$

$$d(f_n(x), f(x)) < \varepsilon.$$

→ Uniform conv. implies pointwise conv.

→ Example that converges pointwise but not uniformly

$$f_n : [0, 1] \rightarrow \mathbb{R}$$



Then $\forall x$, $f_n(x) = 0$ for large enough n

but this $\swarrow$
choice of n depends on x
hence it is not uniform

→ If each f_n continuous and $f_n \xrightarrow{\text{uniform}} f$ then f is continuous

Proof

$\epsilon/3$ argument.

① $\frac{\epsilon}{3}$ from continuity of f_n

② $\frac{\epsilon}{3}$ from uniform convergence at x

③ $\frac{\epsilon}{3}$ from uniform convergence at y



Use Triangle Inequality.

→ By contrapositive, discontinuous f 's cannot be approximated by uniformly convergent continuous f_n 's.

→ Let $C(M, N) = \{f: M \rightarrow N \mid f \text{ continuous \& bounded}\}$

be a metric space equipped with sup norm metric

$$d(f, g) = \sup_{x \in M} \{d(f(x), g(x))\}$$

If N is complete, then $C(M, N)$ is complete.

→ Key Insight : Uniform convergence is same as convergence on sup norm

↓
Think about why this is so!

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→ If $f_n \rightarrow f$ uniformly and each f_n continuous then f continuous

→ The space $C(M, \mathbb{R})$ is complete

→ If $f_n \rightarrow f$ uniformly, then

$$\int_a^b f_n(x) dx \longrightarrow \int_a^b f(x) dx$$

$$f_n : [a, b] \rightarrow \mathbb{R}.$$

Proof Let $\varepsilon > 0$. Then $\exists N$ s.t.
 $\forall n \geq N, \quad |f_n(x) - f(x)| < \varepsilon$

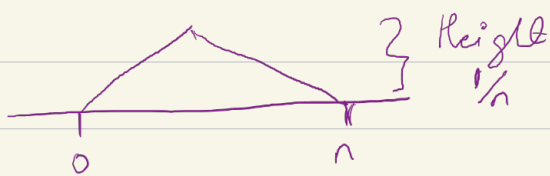
$$\begin{aligned} & \left| \int f_n(x) dx - \int f(x) dx \right| \\ &= \left| \int f_n(x) - f(x) dx \right| \\ &\leq \int |f_n(x) - f(x)| dx < \varepsilon \cdot (b-a) \end{aligned}$$

→ Example with $f_n \rightarrow f$ pointwise
s.t. integrals do not converge

Take the sequence of 'spiky' functions
i.e. $f_n(x)$ is a spike of height
 n from 0 to $1/n$.

Then $f_n \rightarrow 0$ pointwise but the
integrals $\int_0^1 f_n(x) dx = 1/2$ for all n .

→ Example of a ^{sequence of} functions $f_n: \mathbb{R} \rightarrow \mathbb{R}$
s.t. $f_n \xrightarrow{\text{uniformly}} 0$ but $\int f_n \rightarrow 0$



Then $f_n \xrightarrow{\text{uniformly}} 0$

But $\int_0^\infty f_n(x) dx = \frac{1}{2}$
for all n

[For such an example, we need half the
real line - we cannot make such an example
in an interval]

→ If $f_n \Rightarrow f$ and $f_n' \Rightarrow g$

then f is differentiable and $f' = g$.

Proof By FTC,

$$\int_0^x f_n'(y) dy = f_n(x) - f_n(0)$$

↓ Pass to limits on both sides

$$\int_0^x g(y) dy = f(x) - f(0).$$

We are just using convergence of integrals.

→ Power series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

and if we take $R = \liminf_{n \rightarrow \infty} |a_n|^{-1/n}$

Then the power series is absolutely convergent if $|x| < R$.

Take $\delta > 0$ then we claim that

series converges uniformly ^{for} $x \in [-R + \delta, R - \delta]$

$$\left| f(x) - \sum_{n=0}^N a_n x^n \right| = \left| \sum_{n=N+1}^{\infty} a_n x^n \right|$$

$$\leq \sum_{n=N+1}^{\infty} |a_n| |x|^n$$

↑
tail of
the sum

$$\leq \sum_{n=N+1}^{\infty} |a_n| (R-\delta)^n$$

$\searrow \approx R^{-n}$
 for large n

$R-\delta < R$ so we have

$$\left(\frac{R-\delta}{R} \right)^n \quad \text{geometric series}$$

$$\rightarrow \liminf_{n \rightarrow \infty} |a_n|^{-1/n} = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{1/n}}$$

For a monotone decreasing continuous function F ,

$$\liminf_{n \rightarrow \infty} F(x_n) = F\left(\limsup_{n \rightarrow \infty} x_n\right)$$

$$\rightarrow \left(\sum_{n=0}^{\infty} a_n x^n \right)' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

Does this
converge
uniformly?

Candidate is $\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$

So we want to check its radius of convergence which is $\liminf_{n \rightarrow \infty} ((n+1) a_{n+1})^{-1/n} = R$ (*)

Fun fact: $(n+1)^{1/n} \rightarrow 1$ as $n \rightarrow \infty$

Also $\liminf_{n \rightarrow \infty} a_{n+1}^{-1/n} = \liminf_{n \rightarrow \infty} a_n^{-1/n}$

Hence (*) the radius of convergence of derivative is same as that of the original power series

→ Hence we see that power series always gives us C^∞ within the radius of convergence (infinitely differentiable)

→ Continuous functions over an interval are a complete metric space

→ Take $C([a, b])$

with $d(f, g) = \max \{ |f(x) - g(x)| : x \in [a, b] \}$

Claim #1 $C([a, b])$ is complete

Question What are the compact sets of $C([a, b])$?

(Think about totally bounded)

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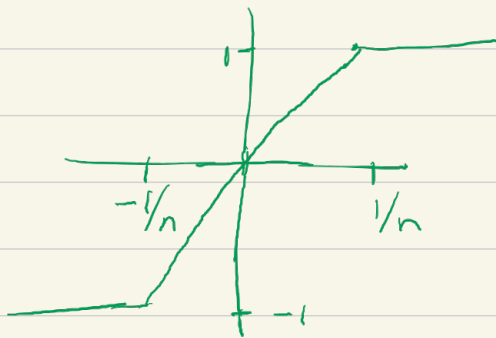
→ $C([a, b], \mathbb{R}) = \{f: [a, b] \rightarrow \mathbb{R} \text{ continuous}\}$

with sup norm

→ Convergence on sup norm is equivalent to uniform convergence

→ If (f_n) is a bounded sequence of functions in $C([a, b])$ there is not necessarily a uniformly convergent subsequence

Example 1



$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Example 2 $f_n(x) = \sin(nx)$

→ A sequence of functions (f_n) is
equicontinuous if $\forall \varepsilon > 0, \exists \delta > 0$
s.t. if
 $d(x, y) < \delta$ then $\forall n$
 $d(f_n(x), f_n(y)) < \varepsilon$.

Here choice of δ is uniform for all
↑ delta functions.

→ Thm Arzela - Ascoli Thm

If $f_n : [a, b] \rightarrow \mathbb{R}$ is a bounded &
equicontinuous seq. of functions then
there exists a uniformly convergent subseq.

* Observe that this is a characterisation of
compact subsets of $C([a, b])$

→ Proof of Arzela Ascoli

Step 1: Find a subseq. that converges pointwise in $[a, b] \cap \mathbb{Q}$.

(Here we only use that $f_n(x)$ is bounded for each $x \in [a, b]$).

Step 2: Use equicontinuity to show the seq. from step 1 actually converges uniformly on $[a, b]$.

→ Proof of Step 1

$[a, b] \cap \mathbb{Q} = \{q_1, q_2, \dots\}$ because countable

Take $f_{n_{1,1}}(q_1), f_{n_{1,2}}(q_1), f_{n_{1,3}}(q_1) \dots$ converges

This is a subseq. of f_n → Here we are using that $f_n(q_1)$ is bounded

Then take a subseq. of the above seq. so that

$$f_{n_{2,1}}(q_2), f_{n_{2,2}}(q_2), f_{n_{2,3}}(q_2) \dots$$

And this is in addition to convergence at q_1 ^{converges}

$\vdots$
So $f_{n_{k,i}}(q_j)$ converges as $i \rightarrow \infty$ for all $j \leq k$.

Each $f_{n_{k,i}}$ is a subseq. of $f_{n_{k-1,i}}$

Diagonal Brilliant Idea

Consider $f_{n_{i,i}}(x)$

→ Proof of Step 2

Let's call f_n the subseq. from part 1
for ease of notation

We know $f_n(x)$ converges pointwise for all $x \in [a, b] \cap \mathbb{Q}$
and f_n is equicontinuous

Proof involves $\frac{\epsilon}{3}$ argument.

Let $\delta > 0$ s.t. $|f_n(x) - f_n(y)| < \frac{\epsilon}{3}$ if $|x - y| < \delta$

Let $q_1, \dots, q_N$ s.t. $\bigcup_{i=1}^N (q_i - \delta, q_i + \delta) \supset [a, b]$

We know $f_n(q_i)$ converges $\forall i \in [N]$

Pick M s.t. if $n, m > M$, $|f_n(q_i) - f_m(q_i)| < \epsilon/3$

Then for $|q_i - x| < \delta$ we have

$$\begin{aligned} |f_n(x) - f_m(x)| &\leq |f_n(x) - f_n(q_i)| \\ &\quad + |f_n(q_i) - f_m(q_i)| \\ &\quad + |f_m(q_i) - f_m(x)|. \end{aligned}$$

→ Bounded slope



equicontinuity



uniformly convergent subseq.



solution to ODE?

→ A set $K \subset C([a, b])$ is compact
iff K is closed, bounded and
equicontinuous

(one direction is Arzela Ascoli - other
direction is proved below)

→ Other direction proof

Let $\epsilon > 0$. Since K is compact,

$$K \subset \bigcup_{i=1}^N B_{\epsilon/3}(f_i)$$

← finite subcover

Each f_i is uniformly continuous
(because continuous on the compact domain $[a, b]$)

hence $\exists \delta_i$ s.t. $|f_i(x) - f_i(y)| < \varepsilon/3$
if $|y - x| < \delta_i$

Pick $\delta = \min \{ \delta_1, \delta_2, \dots, \delta_N \}$

For any $f \in K$, if $|x - y| < \delta$ then

let i be such that $\sup |f_i - f| < \varepsilon/3$

$$\begin{aligned} \text{Then } |f(x) - f(y)| &\leq |f(x) - f_i(x)| \\ &\quad + |f_i(x) - f_i(y)| \\ &\quad + |f_i(y) - f(y)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

$\rightarrow$ A set S of functions $f: M \rightarrow N$ is equicontinuous if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall f \in S$, if $d(x, y) < \delta$ then $d(f(x), f(y)) < \varepsilon$.

$\rightarrow$ Show Arzelà Ascoli fails if domain not compact

$\rightarrow$ ODE
 $x'(t) = f(x(t))$

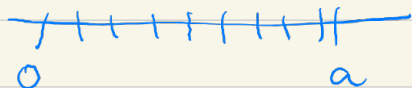
$$x: [a, b] \rightarrow \mathbb{R}$$

$\downarrow$

or say

$$x: [0, a] \rightarrow \mathbb{R}$$

$$x(0) = x_0 \text{ given}$$



$$x_{n+1} = x_n + h f(x_n) = x_n + h x'_n$$

$\rightarrow$ Want to prove that this converges to a solution of the ODE as $h \rightarrow 0$

→ We assume f is bounded & continuous

→ $p_h(x)$ is the polygonal that approximates x with time step h

$$\text{let } h = \frac{a}{N} \quad \text{so } p_{a/N}(0) = x_0$$

$p_{a/N}$ is equicontinuous & bounded.

By Arzel-Ascoli, $\exists$ uniformly convergent subseq.

→ The ODE is equivalent to $\leftarrow$ ^{Dierks} _{Workaround}
$$x(t) = x_0 + \int_0^t f(x(s)) \, ds.$$

Integral form is easier to deal with because uniform convergence passes to integrals (but not derivatives)

→ Above argument shows existence of solution to ODE but not uniqueness

→ Example $f(x) = \begin{cases} \sqrt{x} & \text{for } x \geq 0 \\ 0 & \text{for } x \leq 0 \end{cases}$

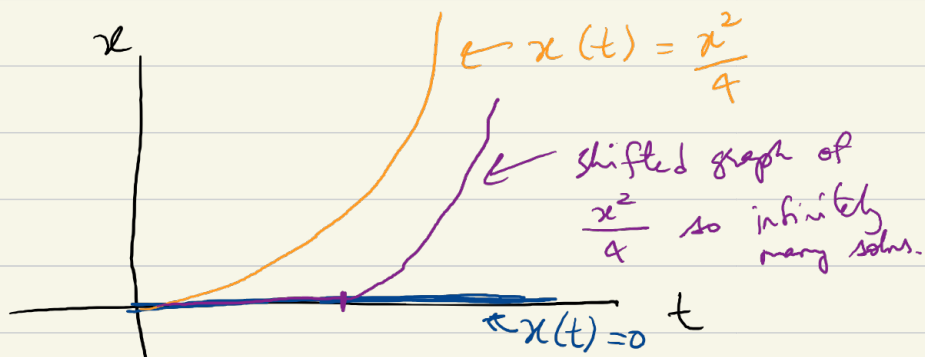
$x'(t) = f(x(t))$ is an ODE (want to find $x(t)$)

One solution is $x(t) = 0$

However $x(t) = \frac{x^2}{4}$ is another soln.

Because $x'(t) = \frac{x}{2}$

and $\sqrt{x(t)} = \frac{x}{2}$



$\rightarrow f: \Omega \rightarrow \mathbb{R}^n$ and $\Omega \subset \mathbb{R}^n$ open
 $x_0 \in \Omega$

Assume f is Lipschitz & continuous then

$\exists a > 0$ s.t. (ODE) has a unique
soln. in $x \in (-a, a) \rightarrow \Omega$

ODE is $x'(t) = f(x(t))$ and $x(0) = x_0$.

→ Contraction Mapping Theorem

$f: M \rightarrow M$ where M is a complete metric space

is a strict contraction if $\exists \lambda < 1$

$$\text{s.t. } d(f(x), f(y)) \leq \lambda d(x, y)$$

→ Then there exists a unique fixed point

$$\text{i.e. } \exists! z \in M \text{ s.t. } f(z) = z$$

→ Shopping centre example

→ Proof

Take a guess for the fixed pt. & iteratively improve

Guess random x_0 and

define $x_{n+1} = f(x_n)$

Then we get a sequence $x_0, x_1, x_2, \dots$

let $l = d(x_0, x_1)$

Then $d(x_1, x_2) = d(f(x_0), f(x_1))$
 $\leq \lambda d(x_0, x_1) = \lambda l$.

So distance is $\leq \lambda^n l$ geometric series

Let m, n be any two (large) indices with
 $m > n$

$$\begin{aligned}
 \text{Then } d(x_m, x_n) &\leq d(x_n, x_{n+1}) \\
 &\quad + d(x_{n+1}, x_{n+2}) \\
 &\quad + \dots \\
 &\quad + d(x_{m-1}, x_m)
 \end{aligned}$$

$$\begin{aligned}
 &\leq \lambda^n l + \lambda^{n+1} l + \dots + \lambda^{m-1} l \\
 &= \frac{\lambda^n l}{1-\lambda}
 \end{aligned}$$

So the sequence $x_0, x_1, \dots$ is Cauchy in a complete metric space so $\rightarrow x^*$ converges

And consider seq. $f(x_0), f(x_1), \dots$ since f is continuous we can pass limits inside

$$\begin{aligned}
 \text{so } x_n &\rightarrow x^* \text{ and } f(x_n) \rightarrow f(x^*) \\
 \text{so } x^* &= f(x^*)
 \end{aligned}$$

Shows existence

For uniqueness - if another fixed pt. y
then $y = f(y)$

$$d(x, y) \leq \lambda d(x, y) \quad \text{only if } d(x, y) = 0.$$

→ Picard Theorem

Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be bounded & Lipschitz

Consider ODE $x'(t) = F(x(t))$ with $x(0) = x_0$

Here $x: \mathbb{R} \rightarrow \mathbb{R}^n$

Picard Thm says that $\exists$ unique soln to the
ODE!

Proof of Picard Thm

→ Observation: ODE is equivalent to integral identity

$$x(t) = x_0 + \int_0^t F(x(s)) \, ds.$$

Think of this as a fixed point problem

$$\rightarrow M = C([-t, t], \mathbb{R}^n)$$

↖
metric
space

$$G(x(t)) = x_0 + \int_0^t F(x(s)) \, ds$$

G is a function that maps curves to curves

$$\text{Want to find } x(t) = (G(x))(t)$$

i.e. ^{want to} apply contraction mapping thm. to G

Given $x(t), y(t) \in C([- \tau, \tau], \mathbb{R}^n)$

$$d(x, y) = \max \{ |x(t) - y(t)| : t \in [- \tau, \tau] \}$$

$$d(G(x), G(y)) = \max \left\{ \int_0^t |F(x(s)) - F(y(s))| ds : t \in [- \tau, \tau] \right\}$$

F is Lipschitz

$$\text{so } |F(x) - F(y)| \leq L|x - y|$$

$$\leq \max \left\{ \int_0^t L|x(s) - y(s)| ds \right\}$$

$$\leq \max \{ L t d(x, y) : t \in [- \tau, \tau] \}$$

$$= L \tau d(x, y) \text{ so want } L \tau < 1$$

$$\text{so define } \tau = \frac{1}{2L}$$

→ We saw characterisation of compact sets in space of continuous functions

What about dense sets in space of continuous functions?

Then Polynomials are dense in $C([a, b])$ (*)

Weierstrass Thm ↓ Generalises

Any algebra of functions that vanishes nowhere and separates points is dense in $C(M)$ where M is a compact metric space.

(*) Proof of this theorem

Want to approximate a continuous function with a polygonal.

A polygonal is a linear combination of tent functions



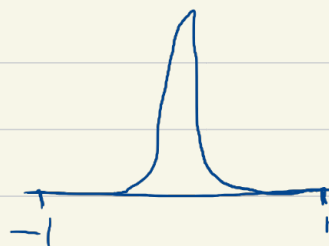
So we would like a function that is non-zero on a small interval and zero elsewhere

(Dirac delta!)

Say on interval $[-1, 1]$, we can define

$$P_m(x) = \overset{\text{constant}}{a_m} \cdot (1-x^2)^m$$

For large m , it looks like



As $m \rightarrow \infty$, $P_m \rightarrow$ Dirac delta δ_x

$$\text{s.t.} \quad \int_{-\infty}^{\infty} \delta_x \, dx = 1$$

$$\text{and} \quad f(x) = \int f(y) \cdot \delta_0(x-y) \, dy$$

$$f * g(x) = \int f(y) g(x-y) dy$$

is called the convolution of f and g

→ Claim If $f \in C([0,1])$ so that $f(0) = f(1) = 0$

$$\text{then } q_m(x) = \int_0^1 f(y) p_m(x-y) dy$$

$$= \int_{\mathbb{R}} f(y) p_m(x-y) dy$$

$$= \int_{-1}^1 p_m(y) f(x-y) dy$$

→ Understand the proof that polynomials are dense in $C([0,1])$

Thm Polynomials are dense in $C([a, b])$ (*)

Weierstrass Thm $\downarrow$ Generalises

Any algebra of functions that vanishes nowhere and separates points is dense in $C(M)$ where M is a compact metric space.

Separates points : $\forall x \neq y \in K$
 $\exists f \in \mathcal{A}$ s.t. $f(x) \neq f(y)$
 $\uparrow$ function $\uparrow$ algebra

Vanishes nowhere : $\forall x \in K, \exists f \in \mathcal{A}$ s.t. $f(x) \neq 0$

$\rightarrow$ Trigonometric polynomials are dense in the space of periodic functions $\rightarrow$ this fact is used to derive Fourier series!

(Is this the space of continuous periodic functions?)

When is second
midterm?

Vector Calculus

→ Set of all linear transformations from one vector space to another is itself a vector space!

$$L(\mathbb{R}^n, \mathbb{R}^m)$$

What about norm in $L(\mathbb{R}^n, \mathbb{R}^m)$?

For $A \in L(\mathbb{R}^n, \mathbb{R}^m)$,

$$|A| = \sup_{x \neq 0} \frac{|Ax|}{|x|} \quad (*)$$

i.e. how large a norm can stretch a non-zero vector

Linear transformations are naturally associated with matrices

$$(*) \quad |A| = \inf \{ \lambda \geq 0 : |Ax| \leq \lambda |x| \}$$
$$= \sup_{|x|=1} \frac{|Ax|}{|x|}$$

are equivalent ways of writing the norm

Given $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\Omega \subset \mathbb{R}^n$
and $x \in \Omega$

$$\text{If } f(x+y) = f(x) + A \cdot y + R(y)$$

$\uparrow$ linear transformation in $L(\mathbb{R}^n, \mathbb{R}^m)$ $\uparrow$ error term

$$\text{and } \lim_{y \rightarrow 0} \frac{|R(y)|}{|y|} = 0$$

We say $Df(x) = A$

where D is the derivative operator

→ 3 November 2025

$f: U \rightarrow \mathbb{R}^m$ with $U \subset \mathbb{R}^n$ open

We say f is diff'able at x
if $\exists A \in L(\mathbb{R}^n, \mathbb{R}^m)$ so
 $\uparrow$
space of
linear
transformation

$$f(x+h) = f(x) + A \cdot h + R(h)$$

$$\text{so } \lim_{h \rightarrow 0} \frac{|R(h)|}{h} = 0$$

i.e. we want $R(h) \ll A \cdot h$

We write $Df_x := A$ as notation

$$f(x+h) = f(x) + Df_x(h) + R(h)$$

$\uparrow$
 Df_x is a linear operator
acting on the vector h

$$U \subset \mathbb{R}^n$$

$f : U \rightarrow \mathbb{R}^m$ can be thought of as n different functions, one for each coordinate

Partial derivatives

$$\frac{\partial f_i}{\partial x_j} := \lim_{h \rightarrow 0} \frac{f_i(x + h e_j) - f_i(x)}{h}$$

$$e_j = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow \begin{matrix} j^{\text{th}} \text{ coordinate is } 1, \\ \text{all others are } 0. \end{matrix}$$

Directional derivative

$$v \in \mathbb{R}^n$$

↑ any arbitrary vector

$$\partial_v f = \lim_{h \rightarrow 0} \frac{f(x + h v) - f(x)}{h}$$

→ Observations

→ If $\partial_v f$ exists for any $v \in \mathbb{R}^n$ then the $\frac{\partial f}{\partial x_i}$ exist and $= \partial_{e_i} f$

→ If Df_x exists then $\partial_v f(x)$ exists and $= Df_x(v)$

↳ Proof

$$\begin{aligned}\partial_v f &= \frac{f(x+hv) - f(x)}{h} = \frac{[f(x) + Df_x(hv) + R(hv)] - f(x)}{h} \\&= \frac{Df_x(hv) + R(hv)}{h} \\&= Df_x(v) + \frac{R(hv)}{h} \\&\quad \downarrow \text{goes to 0 as } h \rightarrow 0 \\&= Df_x(v)\end{aligned}$$

→ But neither of the opposite directions are true

Example

$$f(r\sigma) = r g(\sigma)$$

↑ ↑
norm unit
vector

$$g(\sigma) = -g(-\sigma)$$

$$\text{Then } \partial_\sigma f = g(\sigma)$$

$$\text{Define } f(x, y) = \frac{x^2 \cdot \text{sign}(x)}{\sqrt{x^2 + y^2}}$$

Then check partial derivatives at origin

$$\frac{\partial f}{\partial x}(0, 0) = 1$$

$$\text{and } \frac{\partial f}{\partial y}(0, 0) = 0$$

$$\partial_{(1,1)} f = \lim_{h \rightarrow 0} \left(\frac{h^2 (\text{sign } h)}{\sqrt{2}h^2} - 0 \right)$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{2}} \frac{h}{h} = \frac{1}{\sqrt{2}} \neq 1.$$

Conclusion: Directional derivatives are not a linear combination of the partial derivatives in general

Thm Given $f: U \rightarrow \mathbb{R}^m$ and $x \in U$ $U \subset \mathbb{R}^n$

If $\frac{\partial f}{\partial x_j}$ all exist and are continuous at x

Then DF_x exists

For simpl.

5 Nov '25

→ Chain Rule in multi variable

$$f: U \rightarrow \mathbb{R}^m \quad \text{where } U \subset \mathbb{R}^n$$

$$g: \mathbb{R}^m \rightarrow \mathbb{R}^k$$

Then $x \in U$, consider the composition $g \circ f: U \rightarrow \mathbb{R}^k$

Claim If f is differentiable at x and g is differentiable at $f(x)$ then $g \circ f$ is differentiable at x and $(D(g \circ f))_x = Dg_{f(x)} \cdot Df_x$

$$\downarrow \\ \mathbb{R}^{k \times n}$$

$$\downarrow \\ \mathbb{R}^{k \times m}$$

$$\downarrow \\ \mathbb{R}^{m \times n}$$

→¹⁵ $f: \mathbb{R}^n \rightarrow \mathbb{R}$ then gradient

$$\nabla f = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix}$$

$$d_v f(x) = Df_x(v) = \nabla f(x) \cdot v \quad \begin{array}{l} \swarrow \text{dot product} \\ \cdot v \end{array}$$

If $\gamma : [0, 1] \rightarrow \mathbb{R}^n$ is a curve,

then

$$\frac{d}{dt} (f(\gamma(t))) = \nabla f(\gamma(t)) \cdot \gamma'(t)$$

∇f is the direction of maximum possible growth

$-\nabla f$ is direction of downhill fall!

[Stochastic gradient descent/flow!]

$$\rightarrow f(x, y) = x^2 - y^2$$

Gradient $\nabla f \perp$ all level sets
 $\uparrow$
 perpendicular
 $\perp$

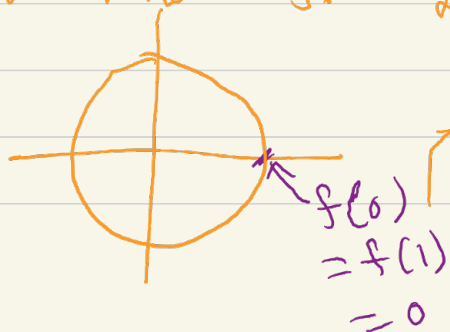
Definition : Level set is a curve that shows all points for which the function has a constant value (i.e. function value doesn't change along a level set).

∇f (gradient) shows direction of maximal growth so we expect it to be orthogonal to the direction where f doesn't change at all, i.e., orthogonal to the level set.

→ Implicit Function Thm

$\exists \gamma$ ^{curve} that characterises a level set.

→ MVT applies in partial derivatives to each component of f individually, but not to the whole f . E.g.



$$\text{so } f(1) - f(0) = 0$$

but derivative $\neq 0$ at any intermediate point

$$\rightarrow f(b) - f(a) = \int_0^1 Df_{(a+t(b-a))} (b-a) dt$$

$$|f(b) - f(a)| \leq \max \|Df\| \cdot |b-a|$$

→ Second Derivatives

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

If $\frac{\partial^2 f_k}{\partial x_i \partial x_j}$ are continuous then

second derivatives exist then the order of partials doesn't matter $\rightarrow$

$$\frac{\partial^2 f_k}{\partial x_i \partial x_j} = \frac{\partial^2 f_k}{\partial x_j \partial x_i}$$

→ 7 Nov

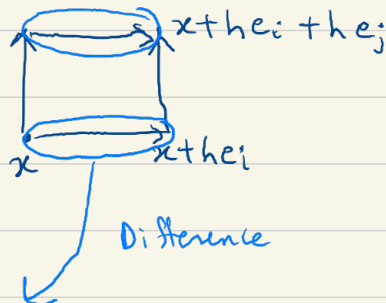
$$f: U \rightarrow \mathbb{R}^m$$

$$Df: U \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$$

$$D^2f: U \rightarrow \mathcal{L}(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$$

→ Claim: order of partials doesn't matter

$$\frac{\partial^2 f_k}{\partial x_i \partial x_j} = \frac{\partial^2 f_k}{\partial x_j \partial x_i}$$



$$\begin{aligned} \textcircled{1} \quad & f(x + h e_i) - f(x) \\ &= \int_0^h \frac{\partial f}{\partial x_i}(x + t e_i) dt \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad & f(x + h e_i + h e_j) - f(x + h e_j) \\ &= \int_0^h \frac{\partial f}{\partial x_i}(x + t e_i + h e_j) dt \end{aligned}$$

Now if I take (2) - (1), I get

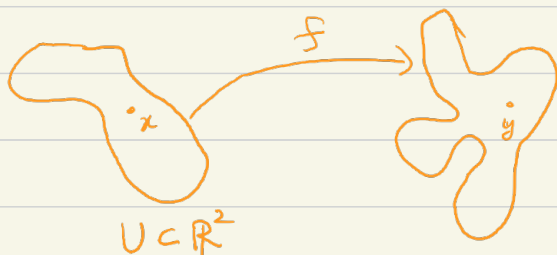
$$\begin{aligned} & f(x + he_i + te_j) - f(x + he_j) - f(x + te_i) + f(x) \\ &= \int_0^h \frac{\partial f}{\partial x_i} (x + he_j + te_i) - \frac{\partial f}{\partial x_i} (x + te_i) dt \\ &= \int_0^h \int_0^h \frac{\partial^2 f}{\partial x_i \partial x_j} (x + te_i + se_j) ds dt \end{aligned}$$

→ Observe that if we change order of partials, we get the same expression

→ Implicit & Inverse Function Theorem

Inverse

$$f: U \rightarrow \mathbb{R}^n, \quad U \subset \mathbb{R}^n$$



Qs: Can we find $g = f^{-1}$?

$$g(f(x)) = x$$

so $g \circ f = \text{Identity}$

$$D(g \circ f)_x = I = (Dg)_y \cdot Df_x$$

$\downarrow$
 Df_x must be
invertible!

Inverse Function Thm

Let $f: U \rightarrow \mathbb{R}^n$ be a C^1 function such
that Df_x is an invertible $n \times n$ matrix

Then $\exists$ a C^1 inverse from a ball around $f(x)$
to a ball around x .

Note: The inverse is local

Proof

WLOG $x=0$ and $f(x)=0$.

① If y is small, $\exists! x \in B_\delta(0)$ s.t. $f(x)=y$

$$\begin{aligned} f(x) &= f(0) + Df_0 \cdot y + R(y) \\ &= \downarrow + Df_0 \cdot y + R(y) \\ &= 0 + Df_0 \cdot y + R(y) \end{aligned}$$

$$x = (Df_0^{-1}) y - (Df_0^{-1}) R(x)$$

Define $m(x) = Df_0^{-1} y - (Df_0^{-1}) R(x)$

Claim If $|y|, |x|$ are small then m is
a contraction so ^{has} unique fixed pt!

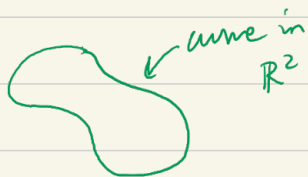
Inverse Function Theorem

$f: \Omega \rightarrow \mathbb{R}^n$ $\Omega \subset \mathbb{R}^n$ open
s.t. $f \in C^1(\Omega)$ and for some
 $x_0 \in \Omega$, $Df(x_0)$ is invertible

Then $\exists \overset{\text{open}}{U} \ni x_0$, $\exists \overset{\text{open}}{V} \ni f(x_0) = y_0$ s.t.
 f is invertible from U to V and $f^{-1}: V \rightarrow U$ is
differentiable

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→ Implicit Function Thm



Two ways to get a curve —

① Level set $\{x \in \mathbb{R}^2 : f(x) = c\}$
usually gives a curve

② Parametrisation $\gamma : [a, b] \rightarrow \mathbb{R}^2$

E.g. unit circle —

① $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a level set
② $\gamma : [0, 2\pi] \rightarrow \mathbb{R}^2$, $\gamma(t) = (\cos t, \sin t)$ definition

→ $f : \mathcal{L} \rightarrow \mathbb{R}^m$

$$f(x_0, y_0) = z_0$$

$$x_0 \in \mathbb{R}^{n-m}, y_0 \in \mathbb{R}^m$$

$\mathcal{L} \subset \mathbb{R}^n$, $n > m$

f is a C^1 in $\mathcal{L}$

very important assumption!

Assume $\frac{\partial f}{\partial y}$ is invertible

U, V open
 $U \subset \mathbb{R}^{n-m}$, $V \subset \mathbb{R}^m$

Then $\exists g : U \rightarrow V$ s.t. $x_0 \in U$, $y_0 \in V$

and $\forall x \in U$, $f(x, g(x)) = z_0$

Pulnam
register

→ Auxiliary function

$$h: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$h(x, y) = (x, f(x, y)) \in \mathbb{R}^n$$

f is C^1 hence h is also C^1

$$h(x_0, y_0) = (x_0, z_0)$$

→ Goal: Apply inverse function theorem to h so that

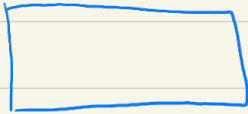
$$h^{-1}(x, z) = (x, y)$$

$$Dh(x_0, y_0) = \left(\begin{array}{c|c} I_{(n-m) \times (n-m)} & 0_{(n-m) \times m} \\ \hline \frac{\partial f}{\partial x} \quad m \times (n-m) & \frac{\partial f}{\partial y} \quad m \times m \end{array} \right)$$

Then Dh is invertible iff $\frac{\partial f}{\partial y} \quad m \times m$ is invertible

Matrix of ^{partial} derivatives is the total derivative matrix
(Jacobian matrix?)

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rectangle
 R

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$\int_R f$$

$$\int_R f(x, y) \, dx \, dy$$

$$\text{Say } R = [a, b] \times [c, d]$$

$$\begin{aligned} \text{Partitions: } \quad a &= x_0 < x_1 < \dots < x_n = b \\ c &= y_0 < y_1 < y_2 < \dots < y_m = d \end{aligned}$$

$$\text{Riemann sum} \quad \int_{[a, b] \times [c, d]} f \, dx \, dy$$

$$= \sum f(x_i, y_j) |R_{ij}|$$

any sampling point
(between m_{ij} and M_{ij})
 $\uparrow_{\text{inf}} \quad \uparrow_{\text{sup}}$

$$R_{ij} = [x_i, x_{i+1}] \times [y_j, y_{j+1}]$$

If Q is a refinement of P , then $L(f, P) \leq L(f, Q)$
 $U(f, P) \geq U(f, Q)$

Proposition If P, Q are any partitions of R ,
 $L(f, P) \leq U(f, Q)$.

Proof Let $S = P \cup Q$ be a common refinement

Then $L(f, P) \leq L(f, S) \leq U(f, S) \leq U(f, Q)$ $\square$

$$\rightarrow \text{If } \sup_P L(f, P) = \inf_P U(f, P) \stackrel{\text{def.}}{=} I$$

then we say f is integrable and

$$\int_R f = I$$

$\rightarrow$ Defn A zero set S is such that
 $\forall \varepsilon > 0, \exists$ countable cover of S by
open rectangles
 $S \subset \bigcup_{i=1}^{\infty} R_i$ s.t. $\sum |R_i| < \varepsilon$.

$\rightarrow$ Fact (Same as single variable)
 f is integrable iff discontinuity $\underset{\text{set of points}}$ is a zero set

$$\text{If } S \subset \mathbb{R}^2,$$

$$|S| = \int \mathbb{1}_S(x, y) \, dx \, dy$$

$$\uparrow_{\text{area}} = \begin{cases} 1 & \text{if } (x, y) \in S \\ 0 & \text{if } (x, y) \notin S \end{cases}$$

Discontinuity points are exactly boundary points

So $\mathbb{1}_S$ is integrable iff
set of boundary points is a zero set

$$\rightarrow f: \overbrace{[a, b] \times [c, d]}^{\mathbb{R}} \rightarrow \mathbb{R}$$

$$\int_{\mathbb{R}} f(x, y) \, dx \, dy \stackrel{?}{=} \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy$$

$$\stackrel{?}{=} \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx$$

$\rightarrow$ In the general case what if
 $\sup L \neq \inf U$?

let us write $\underline{\int} f = \sup_P L(f, P)$ and $\overline{\int} f = \inf_P U(f, P)$

Fubini's
→ Thm

If $\int_R f \, dx \, dy$ exists then

$$\begin{aligned}\int_R f(x, y) \, dx \, dy &= \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy \\ &= \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy\end{aligned}$$

→ Let P be some partition of R

$$P_1 = a = x_0 < x_1 < \dots < x_n = b$$

$$P_2 = c = y_0 < y_1 < \dots < y_m = d$$

$$L(f, R) \leq L\left(\int_a^b f \, dx, P_2\right) \quad \Leftarrow \text{Understand this!}$$

$$\text{So } L(f, P) \leq L\left(\int_a^b f \, dx, P_2\right)$$

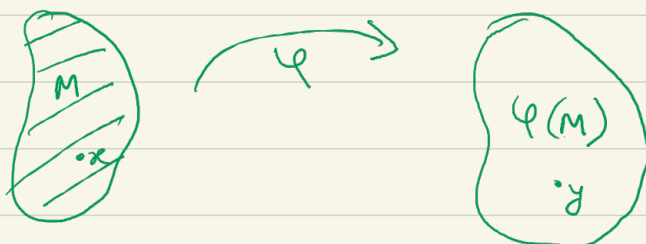
$$\leq U\left(\int_a^b f \, dx, P_2\right)$$

$$\leq U\left(\int_a^b f \, dx, P_2\right)$$

$$\leq U(f, P).$$

→ Change of Variables

$$\int_M f \, dx \, dy$$



$$\int_M f(\varphi(x)) \, |\det D\varphi| \, dx = \int_{\varphi(M)} f(y) \, dy$$

Jacobian

intuition is that
det is the
"area scaling factor"
and $D\varphi$ is the best
linear approximation of φ .

$$f: \varphi(x) \rightarrow \mathbb{R}$$

Suppose φ is bijective from M to $\varphi(M)$

and φ is C^1 so that $D\varphi$ well-defined

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$$\int_D f(\varphi(x)) |\det D\varphi| dx = \int_{\varphi(D)} f(y) dy$$

→ What if we want to integrate
not on volumes but on surfaces or
curves?



partition on
the curve

$\gamma: [a, b] \rightarrow \mathbb{R}^n$ ← parameterization

$$\begin{aligned} \int_C f ds &= \sum_{i=1}^m f(y_i) |x_i - x_{i-1}| \\ &= \sum_{i=0}^m f(\gamma(t_i^*)) |\gamma(t_i) - \gamma(t_{i-1})| \\ &= \int_a^b f(\gamma(t)) \cdot \underbrace{|\gamma'(t)|}_{ds} dt \end{aligned}$$

$$\frac{d}{dt} f(\gamma(t)) = \nabla f \cdot \gamma'(t)$$

$$\begin{aligned} \text{So } \int_a^b \frac{d}{dt} f(\gamma(t)) dt &= f(\gamma(b)) - f(\gamma(a)) \\ &= \int_a^b \nabla f(\gamma(t)) \cdot \gamma'(t) dt \end{aligned}$$

When is $dt = ds$? $\Rightarrow$ It is when $|\gamma'(t)| = 1$.

$$\int_c \nabla f \cdot \tau ds = f(\text{end}) - f(\text{start})$$

$\nwarrow$ dot product
 $\nwarrow$ tangent vector to the curve with unit length

$\rightarrow$ Given a vector field $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ and
a curve $\gamma: [a, b] \rightarrow \mathbb{R}^n$

$$\begin{aligned} \text{The line integral is } \int_c F \cdot \tau ds &\leftarrow \begin{array}{l} \uparrow \text{unit tangent} \\ \text{differential of length} \end{array} \\ &= \int_a^b F(\gamma(t)) \cdot \gamma'(t) dt \end{aligned}$$

Fact: value does not depend on the parametrization

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- ① $\exists P: \mathcal{L} \rightarrow \mathbb{R}^n$ s.t. $F = \nabla P$
- ② $\int_{\text{curve}} F \cdot \tau \, ds$ depends only on endpoints
- ③ For any loop in $\mathcal{L}$, $\int_{\text{loop}} F \cdot \tau \, ds = 0$
- ④ DF is a symmetric matrix

1 December 2025

k -form is something we integrate on surface of dimension k

Def

k -cell given by parametrization

$$\varphi : [0, 1]^k \rightarrow \mathbb{R}^n$$

is a C^1 function



A k -form

$$\omega = \sum_I f_I(x) dx_I$$

$$\int_{\varphi} f_I dx_I = \int_{[0, 1]^k} f(\varphi(u)) \det\left(\frac{\partial(\varphi_{i_1}, \dots, \varphi_{i_k})}{\partial u}\right) du$$

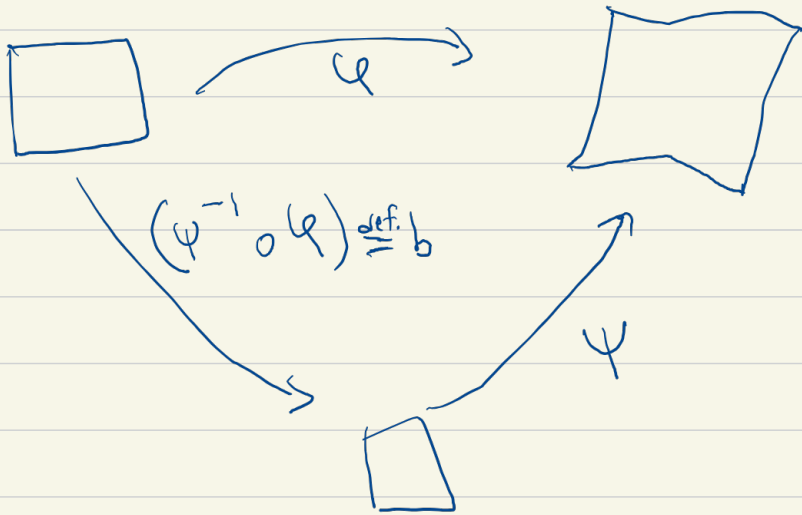
Diagram illustrating the change of variables in the integral:

- An arrow points from the label " k -cell" to the symbol φ in the integral.
- An arrow points from the label " k -form" to the symbol dx_I in the integral.

Proposition

The value of $\int_{\text{surface}} \omega$ does

not depend on the parametrisation
(except for change in orientation)



ψ and φ are 2 different parametrisations

Note that $\varphi = \psi \circ b$

$$\begin{aligned} \int f_I dx_I &= \int f_I (\varphi(u)) \det \left(\frac{\partial \varphi_I}{\partial u} \right) du \\ &\stackrel{(?)}{=} \int f_I (\psi(v)) \det \left(\frac{\partial \psi_I}{\partial v} \right) dv \end{aligned}$$

$$f(\varphi(u)) = f(\varphi(b(u)))$$

$$\det \left(\frac{\partial \varphi_I}{\partial u} \right) = \det \left(\frac{\partial \varphi'_I}{\partial v} \right) \cdot \det(Db)$$

$$dv = \det(Db) \cdot du$$

Observe: If $I = (i_1, i_2, \dots, i_k)$ has a repeated index
then $dx_I = 0$

because the matrix $\frac{\partial \varphi}{\partial u}$ has a repeated row

$$\text{so } \det \left(\frac{\partial \varphi}{\partial u} \right) = 0.$$

A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a 0-form

Wedge product is a way to manipulate differential forms to create higher-order differential forms

$$\text{Exterior derivative } d\omega = \sum_I df_I \wedge dx_I$$

k -form is something you integrate over a k -dimensional surface

$\varphi : [0, 1]^k \rightarrow \mathbb{R}^n$ is a k -cell

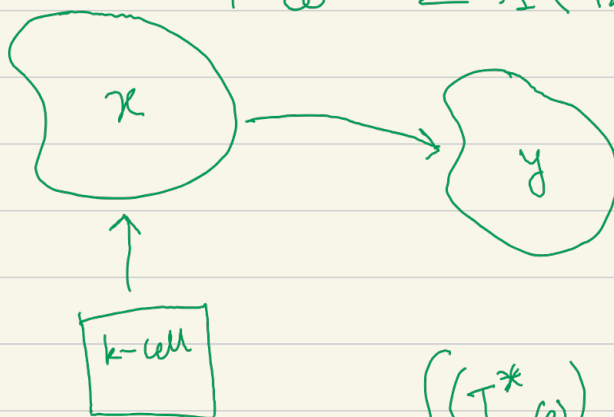
$$\omega = \sum f_I dx_I$$

$$\int_{\varphi} \omega = \sum_I \int_{[0,1]^k} f_I(\varphi(u)) \det(D\varphi_I) du$$

Pull back $\omega = \sum f_I dy_I$

$$y = Tx$$

$$T^* \omega = \sum f_I(Tx) \det\left(\frac{\partial (Tx)_I}{\partial x_J}\right) dx_J$$



$$\int_{\varphi} (T^* \omega) = \int_{\text{Top}} \omega$$

Differential form is a function that assigns a multilinear form

ω is a k -form in $\mathbb{R}^n$

if $\forall x \in \mathbb{R}^n : \omega(x) (v_1, \dots, v_k) \mapsto \text{some real number}$

- multilinear
- alternate

$\uparrow$
 k vectors
in $\mathbb{R}^n$

$$dx_I (v_1, \dots, v_k) = \det \begin{pmatrix} (v_1)_I & (v_2)_I & \dots & (v_k)_I \end{pmatrix}$$

$\uparrow$
 k vectors
in $\mathbb{R}^n$

Space of alternate form is a finite dimensional vector space with basis dx_I 's

Stokes Thm

$$\int_S \partial \omega = \int_{\partial S} \omega$$

$\uparrow$
(k+1)
surface

$\uparrow$
boundary